

ABC's of Estimating Equations



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Acknowledgements

Funding: K01AI177102 (PNZ), K99DA061935 (RKR),
R01AI157758 (PNZ)

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government

A special thank you to Bonnie Shook-Sa, whose prior slides for this
section inspired the current presentation



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Section 1: introduction

Break (15min)

Section 2: applied examples

Break (15min)

Section 3: in context

Statistical Properties

Why estimating equations are useful comes from their statistical properties

- Review the statistical theory
- Illustrate how they apply

We want our estimators to work ‘well’

- But what does ‘well’ mean?

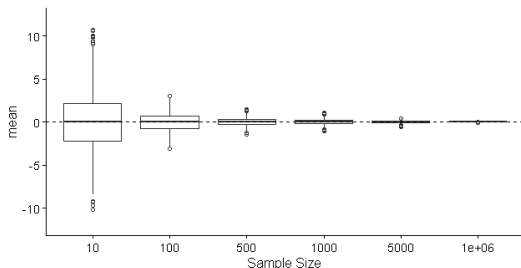
Here, we take a frequentist perspective¹

- Want estimator to be *consistent* and *asymptotically normal*

¹or a large sample framework. Further details on the statistical framework used here are also reviewed in Shook-Sa et al. *Epidemiologic Methods* 2026; 15(1)

Definition: as the number of observations continues to increase, the estimator becomes arbitrarily close (ϵ) to the population parameter θ_0

$$\lim_{n \rightarrow \infty} \Pr(|\hat{\theta} - \theta_0| > \epsilon) = 0$$

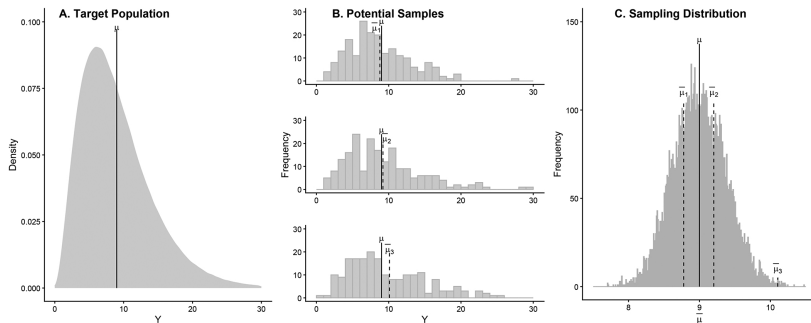


This is why we trust the point estimate of the estimator²

²When we also trust the underlying assumptions

Asymptotic Normality

Definition: as the number of observations continues to increase, the distribution for the estimator converges to a normal distribution³



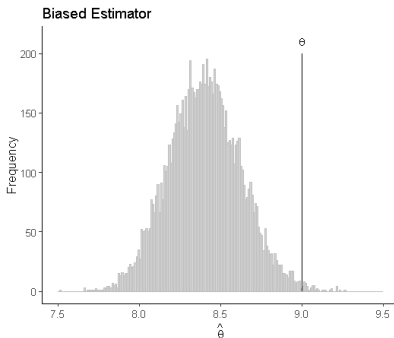
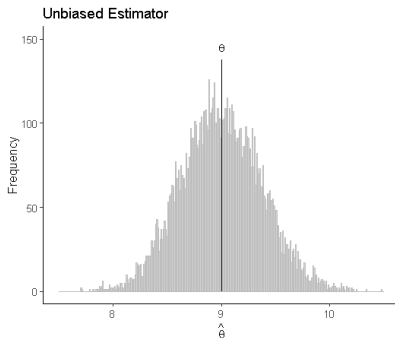
³Figure 1 from Shook-Sa et al. *Epidemiologic Methods* 2026

Asymptotic normality is important for us because it is what justified our use of the sandwich variance and then Wald-type confidence intervals

- Otherwise, the sandwich variance and Wald-type confidence intervals would not be appropriate

Definition: how wrong the estimator is on average

- $E[\hat{\theta}] - \theta_0$



If we can show that an estimating equation is unbiased for θ_0 ,⁴ then it follows that the corresponding estimator is consistent and asymptotic normality

Formally,

$$\sqrt{n}(\hat{\theta} - \theta_0) \rightsquigarrow_d \text{Normal}(0, V(\theta_0))$$

where $V(\theta_0)$ is the sandwich variance

So, our estimator is statistically justified if we can show it is unbiased

⁴...and some suitable regularity conditions, like a unique solution, the estimating function is differentiable, the bread matrix is invertible. See Boos & Stefanski 2013

Goal: show that

$$E[\psi(O_i; \theta_0)] = 0$$

for the estimating function

$$\psi(O_i; \theta) = \begin{bmatrix} R_i(Y_i^* - \mu\alpha) \\ (1 - R_i)(Y_i^* - \alpha) \end{bmatrix}$$

Strategy: divide and conquer

Starting with α : From the law of total expectation

$$E[(1 - R_i)(Y_i^* - \alpha_0)] = E[Y_i^* - \alpha_0 \mid R_i = 0] E[1 - R_i]$$

Then

$$\begin{aligned} E[Y_i^* - \alpha_0 \mid R_i = 0] &= E[Y_i^* \mid R_i = 0] - \alpha_0 \\ &= \alpha_0 - \alpha_0 = 0 \end{aligned}$$

following from linearity and the definition of α

From the law of total expectation

$$E[R_i(Y_i^* - \mu_0\alpha)] = E[Y_i^* - \mu_0\alpha \mid R_i = 1] E[R_i]$$

and then focusing on

$$\begin{aligned} E[Y_i^* - \mu_0\alpha \mid R_i = 1] &= E[Y_i^* \mid R_i = 1] - \mu_0\alpha \\ &= E[Y_i^* \mid R_i = 1] - \mu^* \\ &= \mu^* - \mu^* = 0 \end{aligned}$$

following from linearity of expectations, $\mu = \mu^*/\alpha$, definition of μ^*

Proof from Part 1

Having shown that each of the estimating equations is unbiased at $\theta_0 = (\mu_0, \alpha_0)$, it follows that our estimator is consistent and asymptotically normal

This results is what justifies the usage in Part 1

This process can be repeated for the estimators in Part 2

- But there are other strategies for proving unbiased estimating equations a bit easier than the algebra done for Part 1

Limitations

One important regularity condition is there is a unique solution

This rules out certain estimators

- Most notably very flexible estimators, like neural networks or random forests

So not all estimators can be expressed as estimating equations

Restrictions on Estimating Functions

As suggested, not all estimating functions are *valid*⁵

- Should only depend on the observation i
 - Ex: Cox proportional hazard model
- Should be differentiable at θ_0
 - Ex: median

Need additional criteria or tricks in these cases

⁵See Stefanski & Boos *The American Statistician* 2002; 56(1):29-38 for further details on these conditions

Based on large sample theory

Sandwich variance can perform poorly with small sample sizes

- Finite sample corrections have been proposed⁶

⁶See Saul & Hudgens *Journal of Statistical Software* 2020; 92:1-15 for a review of some approaches

Assumed that θ only has a finite number of parameters

In some *nonparametric* settings, the number of parameters is not finite

- Ex: estimating the survival function, when time is continuous

Previous statistical theory section does not apply anymore

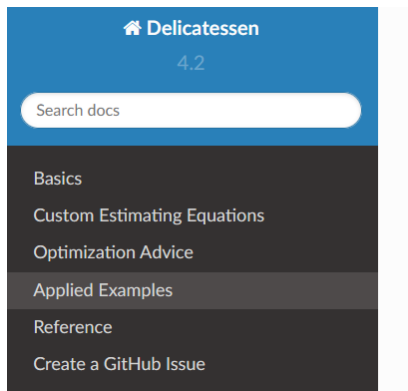
There is theory for when θ is infinite-dimension, but is more complex to use⁷

⁷I am unable to explain how it works yet. But see Kosorok *Introduction to Empirical Processes and Semiparametric Inference* 2008

Next Step(ping Stone)s for your Estimating
Equation Journey

Introduction

- Ross et al. 'M-estimation for common epidemiological measures: introduction and applied examples' *Int J Epidemiol* 2024; 53(2):dyae030
- Cole et al. 'Illustration of 2 fusion designs and estimators' *Am J Epidemiol* 2023; 192(3):467-474
- Boos & Stefanski *Essential Statistical Inference: Theory and Methods* New York: Springer. 2013
- Stefanski & Boos 'The Calculus of M-estimation' *Am Stat* 2002; 56(1):29-38
- Zivich & Lu H 'Constructing g-computation estimators: two case studies in selection bias' *Epidemiol* 2025; 37(1):50-56.



`deli.readthedocs.io`

Notation for 39 different estimating functions

- Regression, causal inference, pharmacokinetic models

28 different applications with publicly-available data

- Missing data, epidemiology, economics, other social sciences, natural sciences

Survival analysis

- Zivich et al. 'Estimating equations for survival analysis with pooled logistic regression' *arXiv:2504.13291*

Time-varying confounding

- Zivich et al. 'Empirical sandwich variance estimator for iterated conditional expectation g-computation' *Stat Med* 2024; 43(29): 5562-5572

Statistical inference on multiple parameters

- Zivich et al. 'Confidence Regions for Multiple Outcomes, Effect Modifiers, and Other Multiple Comparisons' *arXiv:2510.07076*

Open Discussion

Optional Coding Exercise

Choose Your Own Adventure

These are some coding activities to work on yourself using what we have covered. These are exercises meant to help develop your skills in applying estimating equations

1. Adapt the code from Part 2 to estimate the Average Treatment Effect in the Treated (ATT)
2. Adapt the code from Part 2 into an Augmented Inverse Probability Weighting (AIPW) estimator
3. Data Fusion for transporting and measurement error correction

1. Estimate ATT

On GitHub, the standardization and IPW estimators are provided. But try to discern what the equations are based on the following information

- For standardization, we want to compute the mean of the pseudo-potential outcomes *but only among the treated*. How do we restrict contributions to only the 'treated'?
- For IPW, we need to modify the weights. What do the weights look like?⁸

⁸See Sato & Matsuyama 'Marginal structural models as a tool for standardization' *Epidemiol* 2003; 14(6):680-686

2. Estimate the ATE with AIPW

There are a few ways to implement AIPW,⁹ but I will describe a fairly easy variation: weighted-regression AIPW

- Estimate IPTW as before
- Estimate the outcome model, *weighted by IPTW*
- Apply g-computation as before

Estimating equations for weighted regression with weights ω_i

$$\psi(O_i; \beta) = \omega_i(Y_i - \hat{Y}_i)X_i$$

⁹See Shook-Sa et al. *Biometrics* 2025; 81(2):ujaf054 for details. This paper also proves the important aside that this AIPW estimator is doubly-robust for point *and* variance estimation, which isn't the case for most implementations

3. Data Fusion

Go to `deli.readthedocs.io/en/latest/Examples/Cole-AJE-2023.html` and look at Case 3. Recreate the data from the tabled data provided

Try to adapt the code to estimate the measurement-error corrected mean

- Estimating functions are also described using notation in Cole et al. 'Cole et al. Respond to "Combining Information From Diverse Sources"' *Am J Epidemiol* 2024; 193(5):751-752